

Xu-Cheng He

ACADEMY RESEARCH FELLOW / GROUP LEADER

University of Helsinki

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Education

University of Helsinki

Finland

PHD ATMOSPHERIC SCIENCE (DISTINCTION)

2018.01 - 2021.09

- Advisors: Prof. Markku Kulmala, Prof. Jasper Kirkby

University of Helsinki

Finland

MSc ATMOSPHERIC SCIENCE

2015.08 - 2017.10

Yunnan University

China

BSc ATMOSPHERIC SCIENCE

2011.09 - 2015.07

Professional Experience

2025.01-Present	Academy Research Fellow and group leader (2025.09-); Academy post-doctoral researcher and sub-group leader (2025.01-2025.08) , University of Helsinki
2017.09-Present	CERN CLOUD collaboration: Principal Investigator (2026-); Co-Principal Investigator (2024-2026); Coordinator of marine experiments (2017-2023) , CERN / University of Helsinki
2023.01-2024.12	Visiting Researcher (simulation of aerosols) , University of Cambridge
2022.03-2022.12	Visiting Researcher (global model training) , Carnegie Mellon University and Finnish Meteorological Institute

Awards

2025	Airmodus best oral presentation award , The 22nd International Conference on Nucleation and Atmospheric Aerosols	€200
2023	Invited participant of the Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XVII) , Brookhaven National Laboratory	
2022	Extraordinary Potential Prize of 2021 Chinese Government Award for Outstanding Self-financed Students Abroad (20 recipients globally) , Ministry of Education, China	\$10,000
2022	Doctoral dissertation prize, all faculties (4/500) , University of Helsinki	€4,000
2022	NOSA ECS Aerosologist award (Early Career Scientist) , Nordic Society for Aerosol Research	
2021	Outstanding thesis award , Doctoral school in Natural Sciences, University of Helsinki	€2,000
2015	International student grant , University of Helsinki	€1,500
2015	Best thesis award (1/43) , Yunnan University	
2012	Campus star in science and innovation (10/4000) , Yunnan University	¥2,000

Fellowships & Grants

2025-2028	ERC application boost funding , University of Helsinki	€150,000
2025-2029	Academy research fellowship , Research Council of Finland / University of Helsinki	€1,126,320
2025-2026	co-PI, Constraining the size distribution and chlorine production of ferric chloride aerosols for quantitative atmospheric methane removal , Spark Climate Solutions	\$299,248
2022-2025	Postdoctoral researcher fellowship , Research Council of Finland / University of Helsinki	€364,377
2022	Postdoctoral research grant , Jenny and Antti Wihuri foundation	€56,000
2018-2021	Doctoral school fellowship , University of Helsinki	ca. €108,000

Key publications

PUBLICATIONS AS A **CO-FIRST**[%] OR **CORRESPONDING AUTHOR**[#]. NAMES IN BOLD DENOTE CURRENT OR FORMER MEMBERS OF MY RESEARCH GROUP OR COLLABORATORS WHOM I HAVE DIRECTLY MENTORED.

Summary: 6 manuscripts as (co-)first-author and 7 as last-author, in total 12 as corresponding author, including 2 in Science and 2 in Nature.

2026

A. Shcherbinin^{#%}, **J. Shen**^{#%}, O. Kausiala, ..., H. Finkenzeller, M. Rissanen, **X.-C. He**, Networked Open-Source Infrastructure for Scalable Analysis of Chemical Ionization Mass Spectrometry Data. **ChemRxiv preprint**. (2026).

W. Yu, R. Baalbaki, J. Shen, ..., I. Riipinen, **X.-C. He**[#], K. Lehtipalo, Impact of humidity on aerosol growth from methanesulfonic acid. **Environmental Science: Atmosphere**. (2026).

R. Baalbaki[%], **J. Shen**[%], M. Simon, ..., J. Kirkby, K. Lehtipalo[#], **X.-C. He**[#], Role of methanesulfonic acid in atmospheric particle nucleation and growth. **Nature**. (2026).

X.-C. He, N.L. Abraham[#], H. Ding, ..., J. Weber, A.T. Archibald[#], H. Gordon, Evaluation of UKESM aerosol size and composition using ATom measurements indicates missing marine aerosol formation mechanisms. **Atmospheric Chemistry and Physics**. (2026).

2025

R. Zhang, H.-B. Xie[#], F. Ma, R. Yin, J. Chen, **X.-C. He**, Quaternary Nucleation of Iodine and Sulfur Oxoacids in the Marine Atmosphere: Unexpected Role of Methanesulfonic Acid. **Journal of Geophysical Research: Atmosphere**. (2025).

2024

J. Shen[%], D.M. Russell[%], J. DeVivo[%], ..., J. Kirkby[#], J. Curtius[#], **X.-C. He**[#], New particle formation from isoprene under upper tropospheric conditions. **Nature**. (2024).

B. Rörup, X.-C. He[#], J. Shen[%], ..., R. Volkamer, D. Worsnop, K. Lehtipalo, Temperature, humidity, and ionisation effect of iodine oxoacid nucleation. **Environmental Science: Atmosphere**. (2024).

Y. Zhang[%], **D. Li**[%], **X.-C. He**[#], ..., J. Jiang, A. Ding, M. Kulmala, Iodine oxoacids and their roles in sub-3 nanometer particle growth in polluted urban environments. **Atmospheric Chemistry & Physics**. (2024).

2023

X.-C. He[#], M. Simon, S. Iyer, H.-B. Xie[#], ..., N.M. Donahue, M. Sipilä[#], M. Kulmala[#], Iodine oxoacids enhance nucleation of sulfuric acid particles in the atmosphere. **Science**. (2023).

X.-C. He[#], **J. Shen**[#], S. Iyer[%], ..., J. Mikkilä, M. Sipilä, J. Kangasluoma, Characterisation of gaseous iodine species detection using the multi-scheme chemical ionisation inlet 2 with bromide and nitrate chemical ionisation methods. **Atmospheric Measurement Techniques**. (2023).

F. Ma, H.-B. Xie[#], R. Zhang[%], ..., M. Engsvang, J. Elm, **X.-C. He**[#], Enhancement of Atmospheric Nucleation Precursors on Iodic Acid Induced Nucleation: Predictive Model and Mechanism. **Environmental Science and Technology**. (2023).

2022

H. Finkenzeller^{%#}, S. Iyer[%], **X.-C. He**[%], ..., T. Kurten[#], M. Rissanen, R.V. Volkamer[#], The gas-phase formation mechanism of iodic acid as an atmospheric aerosol source. **Nature Chemistry**. (2022).

R. Zhang, H.-B. Xie[#], F. Ma[%], ..., M. Sipilä, M. Kulmala, **X.-C. He**[#], Critical Role of Iodous Acid in Neutral Iodine Oxoacid Nucleation. **Environmental Science & Technology**. 56, 14166-14177 (2022).

2021

M. Wang[%], **X.-C. He**^{%#}, H. Finkenzeller, S. Iyer, D. Chen[%], ..., M. Rissanen, R. Volkamer, Y. J. Tham[#], N. M. Donahue, M. Sipilä, Measurement of iodine species and sulfuric acid using bromide chemical ionization mass spectrometers. **Atmospheric Measurement Techniques**. 14, 4187-4202 (2021).

X.-C. He[#], Y. J. Tham, L. Dada, M. Wang, H. Finkenzeller[%], ..., N. M. Donahue, R. Volkamer, J. Kirkby[#], D. R. Worsnop, M. Sipilä[#], Role of iodine oxoacids in atmospheric aerosol nucleation. **Science**. 371, 589-595 (2021).

X.-C. He[#], S. Iyer, M. Sipilä, A. Ylisirniö, M. Peltola[%], ..., V.-M. Kerminen, R. C. Flagan, J. Kirkby[#], T. Kurtén, M. Kulmala, Determination of the collision rate coefficient between charged iodic acid clusters and iodic acid using the appearance time method. **Aerosol Science & Technology**. 55, 231-242 (2021).

Y. J. Tham, **X.-C. He**, Q. Li, C. A. Cuevas, J. Shen,..., M. Kulmala, C. O'Dowd, M. Dal Maso, A. Saiz-Lopez[#], M. Sipilä[#], Direct field evidence of autocatalytic iodine release from atmospheric aerosol. **Proceedings of the National Academy of Sciences**. 118 (2021).

2020

M. Wang[%], W. Kong[%], R. Marten, **X.-C. He**,..., J.H. Seinfeld, I. El-Haddad, R.C. Flagan, N.M. Donahue[#], Rapid growth of new atmospheric particles by nitric acid and ammonia condensation. **Nature**. 581 (2020).

2019

D. Zhao, R. Yang[#], Y. Tao, W.K. Zhang and **X.-C. He**, Objective detection of the Kunming quasi-stationary front. **Theoretical and Applied Climatology**. 138 (2019).

2017

S. Iyer[#], **X.-C. He**, N. Hyytinen, T. Kurtén[#] and M.P. Rissanen, Computational and Experimental Investigation of the Detection of HO₂ Radical and the Products of Its Reaction with Cyclohexene Ozonolysis Derived RO₂ Radicals by an Iodide-Based Chemical Ionization Mass Spectrometer. **The Journal of Physical Chemistry A**. 121 (2017).

F. Bianchi[#], O. Garmash, **X.-C. He**,..., M. Kulmala, M. Ehn and H. Junninen, The role of highly oxygenated molecules (HOMs) in determining the composition of ambient ions in the boreal forest. **Atmospheric Chemistry and Physics**. 17 (2017).

Total publication count: 72. Total citations: Researchgate (3,394), Google scholar (3,546).

Presentations

SELECTED INVITED TALKS

2025.12 - 2026.01. Invited seminars, **Dalian University of Technology, Nanjing University, Chinese Academy of Sciences**, China

2025.05. Invited seminar, **University of Reading**, UK

2025.03. Invited seminar, **University of Cambridge**, UK

2025.02. Invited seminar, **California Institute of Technology**, USA

2024.06. Invited seminar, **University of Helsinki**, Finland

2024.05. Invited seminar, **Max Planck Institute for Chemistry**, Germany

2024.04. Invited seminar, **Massachusetts Institute of Technology**, USA

2024.04. Invited seminar, **SOLAS open seminar series**

2024.03. Invited seminar, **University of Cambridge**, UK

2024.01. Invited seminar, **University of Birmingham**, UK

2022.08. Invited speaker, **Gordon Research Conference**, Italy

2022.05. Invited talk, **Carnegie Mellon University**, USA

2021.01. Invited talk, **Nanjing University**, China

CONTRIBUTED PRESENTATIONS

American Geophysical Union annual meeting (session convener, 2026)

American Meteorological Society annual meeting (session co-chair, 2026, 2025, 2024, 2023)

Atmospheric Chemistry Colloquium for Emerging Senior Scientists (Invited, 2023)

Gordon Research Conference - Molecular and Ionic Clusters (Invited talk, 2022)

Gordon Research Conference - Atmospheric Chemistry (Poster, 2025, 2023)

European Geosciences Union General Assembly (Talk, 2021, 2020)
 International Aerosol Conference (Talk, 2026, 2023; Poster, 2017)
 European Aerosol Conference (Talk, 2024, 2022; Poster, 2019)
 International Conference on Nucleation and Atmospheric Aerosols (Talk, 2025, 2023; Poster, 2017)
 International Conference on Aerosol Cycle (Talk, 2017)
 Free Radical Symposium (Poster, 2017)
 Surface Ocean Lower Atmosphere Study (Poster, 2019)
 Cryosphere and Atmospheric Chemistry (Poster, 2017)

Teaching Experience

2026	Summer School on Formation and Growth of Atmospheric Aerosols , Lecturer	<i>University of Helsinki</i>
2024	Tropospheric Atmospheric Chemistry , Guest Lecturer	<i>University of Helsinki</i>
2024	Synthesis of physical chemistry, experiments, observations and models to understand atmospheric particle formation and climate impact , Lecturer	<i>University of Helsinki</i>
2019	Climate science at high latitudes: eScience for linking Arctic measurements and modeling , Teaching Assistant	<i>University of Helsinki</i>
2018	Formation and growth of atmospheric aerosols , Teaching Assistant	<i>University of Helsinki</i>

Service, outreach & Professional Development

CONFERENCE CHAIR

2022-2026: Formation and impacts of atmospheric aerosols and cloud condensation nuclei: experiment, observation, and modeling. Co-chair. American Meteorological Society Annual Meeting.
 2025: Young Principal Investigators session. Chair. International Conference on Nucleation and Atmospheric Aerosols.
 2026: Aerosols, clouds and their precursors in the marine atmosphere. Co-chair. International Aerosol Conference 2026.
 2026 - : From Molecules to Climate: Process-Level Understanding of Aerosol and CCN Formation. Convener. American Geophysical Union Annual Meeting.

MANUSCRIPT PEER REVIEW

Science, Nature Geoscience, One Earth, Environmental Science & Technology, EST Letters, EST Air, Geophysical Research Letters, Journal of Geophysical Research: Atmospheres, Atmospheric Chemistry and Physics.

FUNDING REVIEW

National Science Foundation (USA)

SERVICE

2023-present: Coordinator for the modelling group at the CLOUD project, CERN
 2017-present: Run leader for the marine and polar aerosol formation program at the CLOUD project, CERN
 2011-2012: Staff Member, Student Union Office (2011); Deputy Director, Student Union Office (2012)

PUBLIC ENGAGEMENT

2025.10: Delivered a tutorial lecture and guided tour of the CLOUD experiment to a visiting Finnish university delegation.

Supervision & Mentoring

Current group member: Jiali Shen (University Researcher), Dejie Yin (Post-doc), Julia David (Post-doc), Wenjuan Yu (PhD researcher)

PhD supervision/mentoring: Jiali Shen (2023.04 with a distinction), Rima Baalbaki (2024.06), Birte Rörup (2024.06), Wenjuan Yu (on-going)

MSc supervision/mentoring: Ying Zhang (2022.07), Duzitian Li (2024.07)

BSc supervision/mentoring: Duzitian Li (2021.07)

Media Coverage

Dec 2023	Chemistry World , Iodine compounds accelerate cloud formation over oceans and the poles
Oct 2021	Sciencepost , L'émission d'iode par l'océan, une influence inattendue sur la banquise arctique
June 2021	Lab Manager , The Impact of Clouds on Climate Change
Feb 2021	The Atlantic , The Arctic Has a Cloud Problem
Feb 2021	SCIENMAG , Climate research: rapid formation of iodine particles over the Arctic
Feb 2021	PHYS.ORG , CLOUD at CERN reveals the role of iodine acids in atmospheric aerosol formation
Feb 2021	ScienceDaily , How iodine-containing molecules contribute to the formation of atmospheric aerosols, affect climate